🛰️ Derna SAR Flood Monitor

NASA Space Apps Challenge

نظام تجريبي لرصد الفيضانات في مدينة درنة باستخدام تقنية Synthetic Aperture Radar (SAR).

Educational prototype: This version uses simulated SAR data, not actual satellite observations.

📊 Flood Statistics

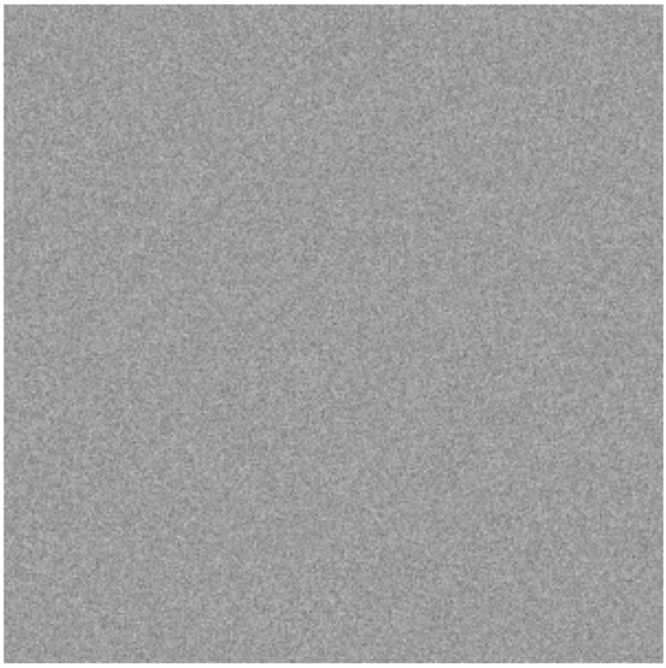
🌊 Affected Area
0.833 km²

📍 Affected Pixels
8,327

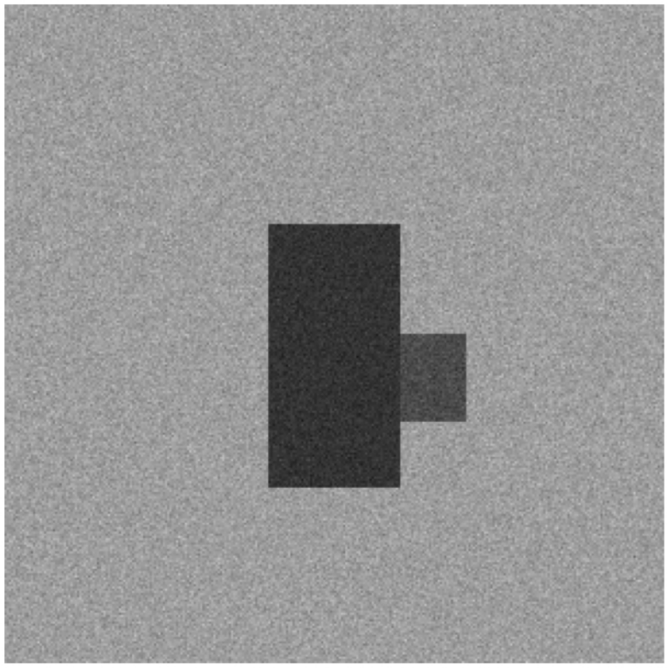
📉 Affected Percentage
9.25%

🛰️ SAR Satellite Images

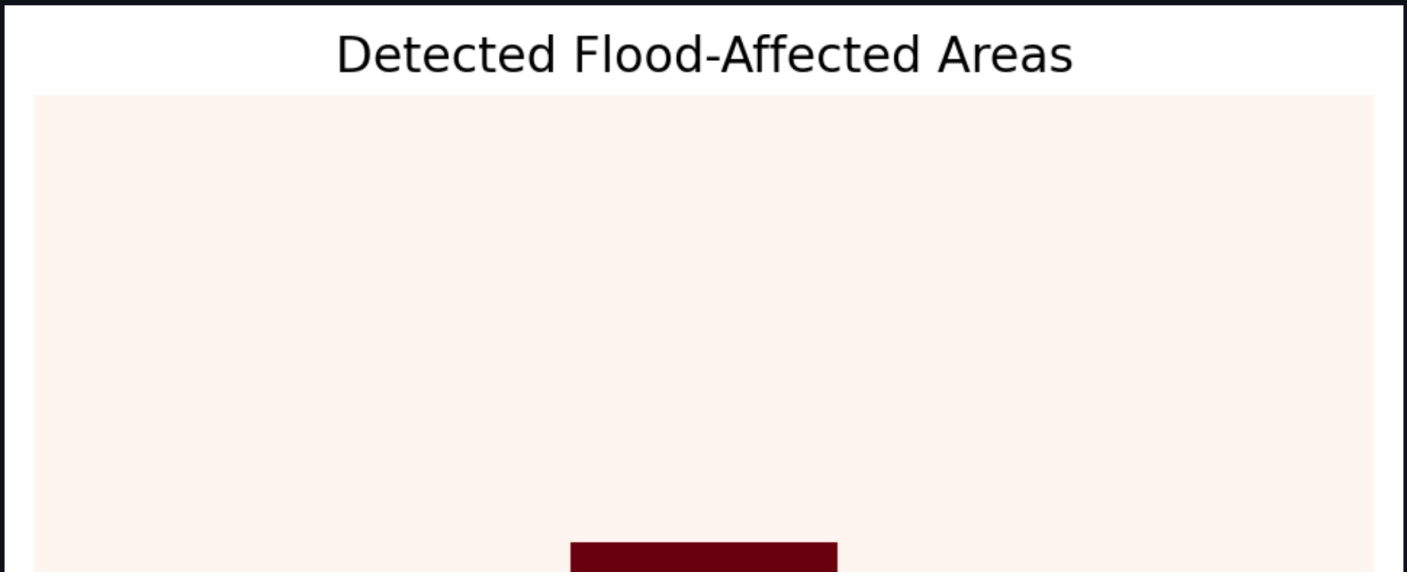
Before Flood - August 2023

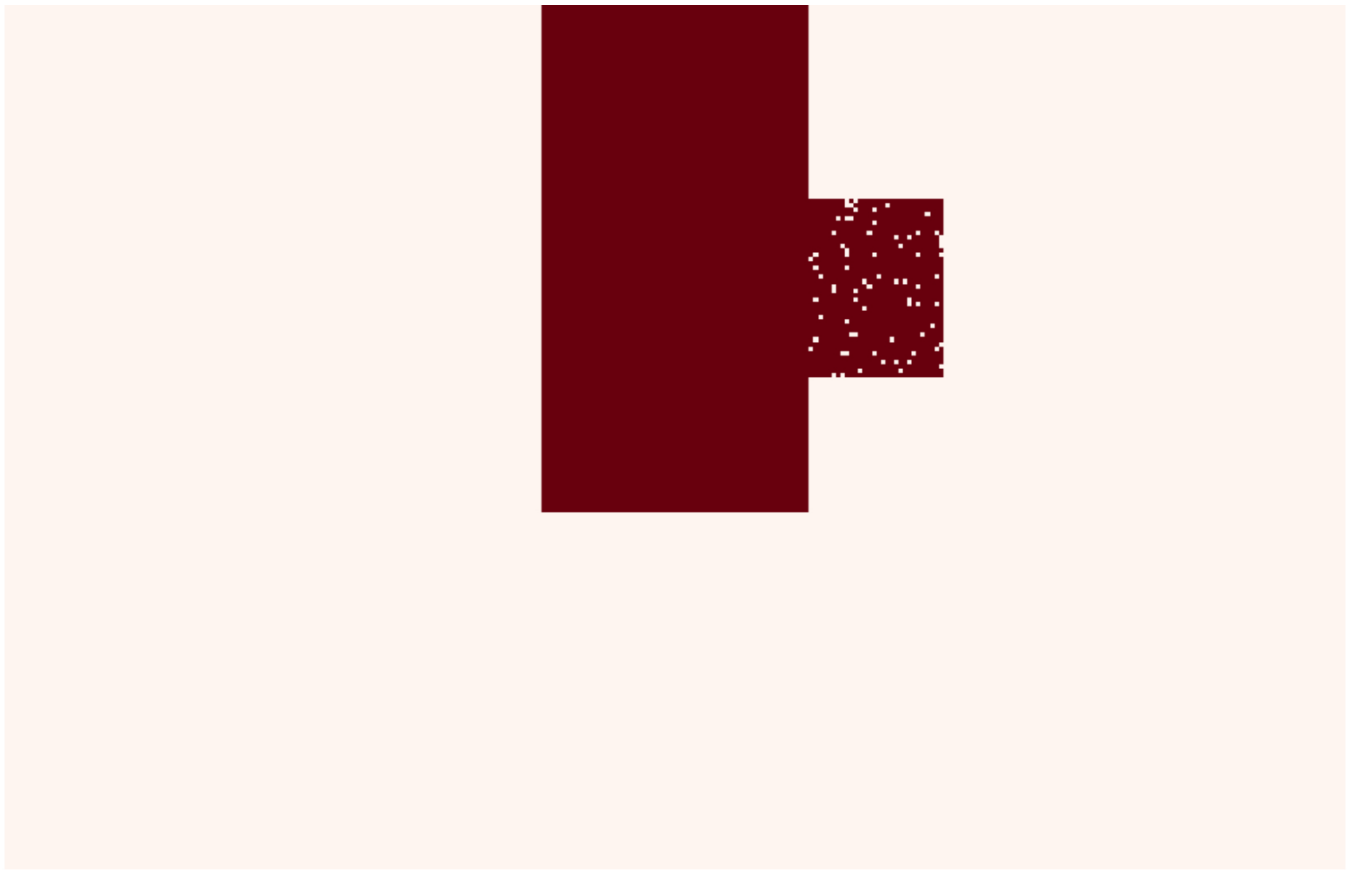


After Flood - September 2023

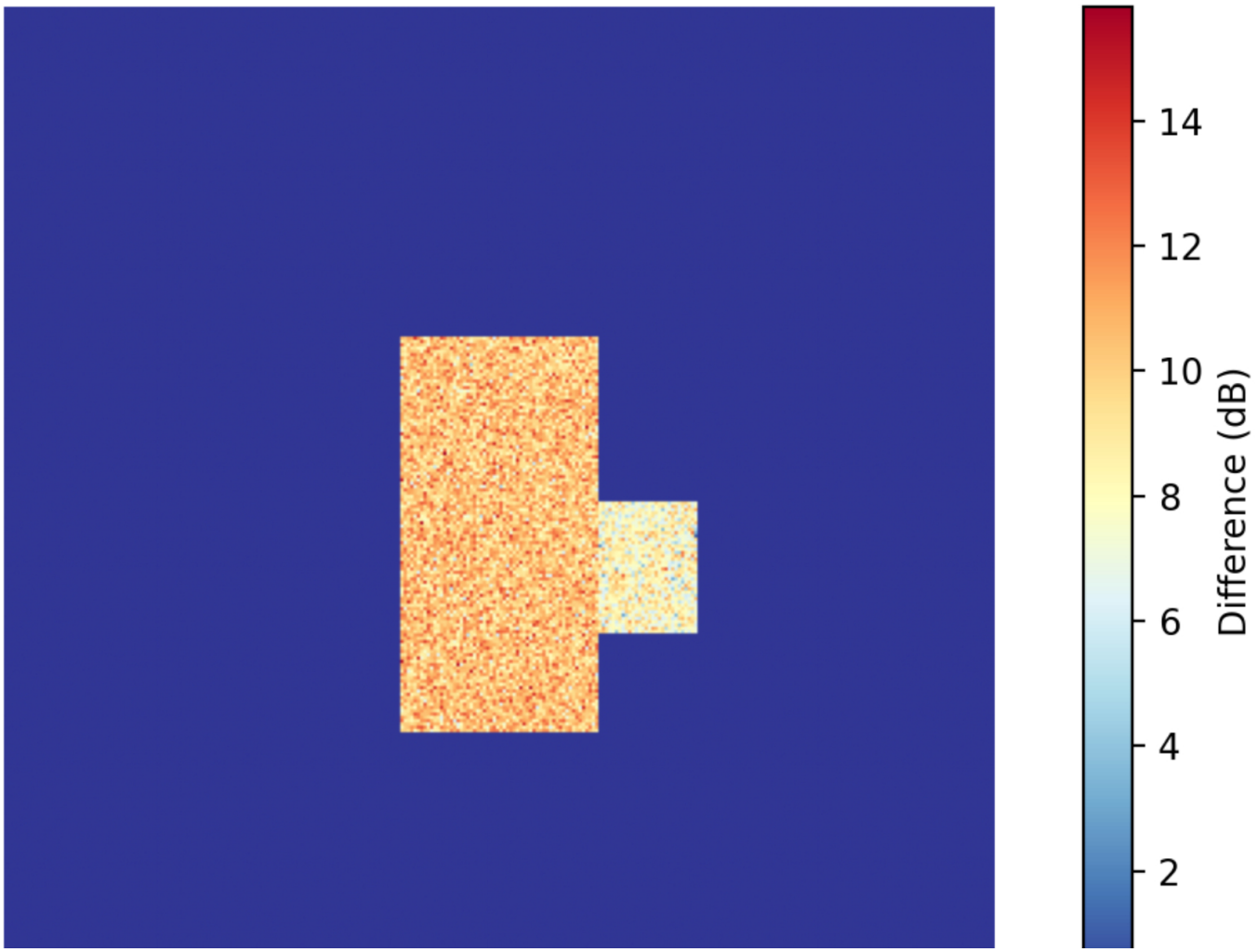


🗺️ Flood Detection Map





Radar Backscatter Difference







📄 Flood Monitoring Report


```
{
  "Project" : "Derna SAR Flood Monitor"
  "Location" : "Derna, Libya"
  "Event" : "September 2023"
  "Data_Type" : "Simulated SAR"
  "Affected_Pixels" : 8327
  "Affected_Area_SqKm" : 0.8327
  "Affected_Percentage" : 9.25
}
```


⬇️ Download JSON Report